

Factors which affect the price of bitcoins

1. Introduction

Bitcoin price does not depend on a **single factor**. It is influenced by a combination of **historical market behavior, technical indicators, macroeconomic conditions, cryptocurrency-specific factors, investor sentiment, supply-side factors, and unexpected news or events**.

Our current prediction model mainly uses historical price and technical-indicator data. Therefore, understanding the other factors is important for explaining why the model sometimes performs well and why it may fail during sudden Bitcoin price movements.

1. Macroeconomic Factors

- **Interest Rates** – Changes in interest rates affect investors' willingness to invest in risky assets such as Bitcoin.
- **Inflation** – Changes in inflation can influence investor demand for Bitcoin.
- **DXY (US Dollar Index)** – Changes in the strength of the US dollar can affect Bitcoin's price.
- **Liquidity / Money Supply** – Higher or lower liquidity in financial markets can influence demand for risk assets.
- **VIX** – Represents market uncertainty and fear and can affect overall risk appetite.
- **S&P 500** – Represents broader stock-market conditions and investor risk sentiment.

2. Crypto-Market Factors

- **Bitcoin ETF Flows** – ETF inflows and outflows indicate institutional buying or selling pressure.

- **Exchange Inflows/Outflows** – Movement of Bitcoin to and from exchanges can provide information about potential buying or selling pressure.
- **Stablecoin Supply** – Changes in stablecoin liquidity can affect the amount of capital available for cryptocurrency trading.
- **Open Interest** – Shows the level of participation and leverage in Bitcoin derivatives markets.
- **Funding Rate** – Indicates the positioning of traders in perpetual futures markets.
- **Liquidations** – Large-scale liquidations can cause sudden increases in buying or selling pressure.

3. On-Chain and Supply Factors

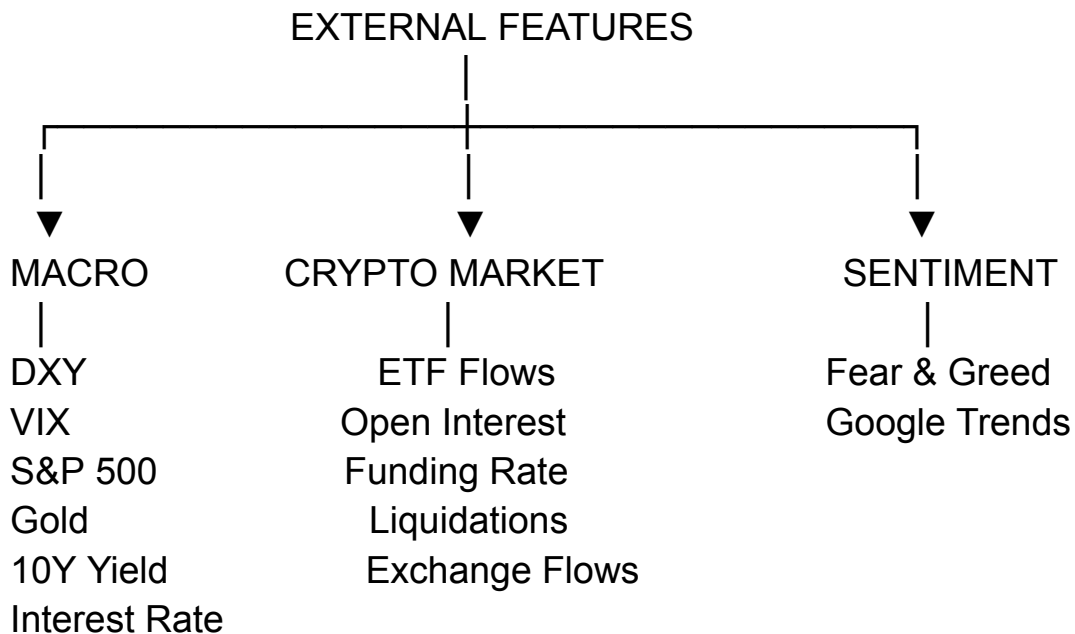
- **Whale Activity** – Large transactions by major Bitcoin holders can influence market supply and sentiment.
- **Exchange Reserves** – The amount of Bitcoin held on exchanges can indicate potential selling pressure.
- **Miner Activity** – Miner selling and reserves can affect Bitcoin's available supply.
- **Hash Rate** – Represents the computational power securing the Bitcoin network.
- **Active Addresses** – Indicates the level of activity and participation within the Bitcoin network.
- **Bitcoin Halving** – Reduces the rate at which new Bitcoin enters circulation and can influence long-term supply dynamics.

4. Investor Sentiment and Information

- **Fear & Greed Index** – Measures overall investor sentiment.
- **News Sentiment** – Positive or negative news can rapidly change market expectations.
- **Regulatory Announcements** – Government policies and regulations can strongly affect Bitcoin demand.
- **Social Media Sentiment** – Public discussions and sentiment on platforms such as X and Reddit can influence short-term market behavior.
- **Google Trends** – Changes in search interest can indicate changes in public attention toward Bitcoin.

But the data is available easily?
Yes, many data easily accessible

Main External features which can improve our model performance



1. Macro Factors

Factor	Meaning	Effect on Bitcoin
DXY	Measures the strength of the US dollar.	DXY ↑ → BTC may ↓; DXY ↓ → BTC may ↑
VIX	Measures fear and uncertainty in financial markets.	VIX ↑ → risk appetite ↓ → BTC may ↓
S&P 500	Represents the overall US stock market.	S&P 500 ↑ → risk appetite ↑ → BTC may ↑
Gold	Represents a major traditional/safe-haven asset.	Changes in gold demand can indicate changes in investor risk preference and may affect BTC demand.

10Y Yield Return investors can earn from US 10-year government bonds.

Yield ↑ → **BTC may become less attractive** → **BTC may** ↓

Interest Rate Cost of borrowing money, mainly influenced by central-bank policy.

Rate ↑ → **risk investment may** ↓ → **BTC may** ↓

2. Crypto-Market Factors

Factor	Meaning	Effect on Bitcoin
ETF Flows	Money entering or leaving Bitcoin ETFs.	Inflow ↑ → buying demand ↑ → BTC may ↑
Open Interest	Total outstanding positions in Bitcoin futures/perpetual markets.	OI ↑ → leverage/market participation ↑ → price movements can become stronger
Funding Rate	Payment between long and short traders in perpetual futures.	Positive → longs dominate ; Negative → shorts dominate
Liquidations	Forced closing of leveraged positions.	Long liquidation → selling → BTC ↓; Short liquidation → buying → BTC ↑
Exchange Flows	BTC moving into or out of cryptocurrency exchanges.	Inflow ↑ → potential selling pressure ↑; Outflow ↑ → available selling supply may ↓

3. Sentiment Factors

Factor	Meaning	Effect on Bitcoin
Fear & Greed	Measures whether investors are fearful or greedy.	Greed → buying pressure ↑; Fear → selling pressure ↑
Google Trends	Measures how much people are searching for Bitcoin.	Search interest ↑ → public attention ↑ → potential demand ↑